

Krishna R. Narayanan, Ph.D.

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Academic Employment and Leadership Roles

2026 – Present	Dean's Fellow for AI , College of Engineering, Texas A&M University
	Associate Department Head for AI , Department of ECE, Texas A&M University
2024 – Present	Sanchez Chair Professor , Department of ECE, Texas A&M University
	Co-Director , Connected Intelligence Initiative, Texas A&M University
2016 – 2024	Eric D. Rubin 'o6 Professor , Department of ECE, Texas A&M University
2022 – Present	Associate Director for Educational Activities , Texas A&M Institute of Data Science
2012 – 2014	Director of Graduate Studies , Department of ECE, Texas A&M University
2008 – Present	Professor , Department of ECE, Texas A&M University
2004 – 2008	Associate Professor , Department of ECE, Texas A&M University
1998 – 2004	Assistant Professor , Department of ECE, Texas A&M University

Visiting Positions

01 – 05/ 2024	Simons Institute for Theory of Computing at Univ. of California at Berkeley
01 – 04/ 2023	Qualcomm Research & Development, San Diego, CA
10 – 11/2022	Massachusetts Institute of Technology, LIDS, Department of EECS
08 – 12/2022	University of Texas at Austin, Department of ECE
01 – 05/ 2015	Simons Institute for Theory of Computing at Univ. of California at Berkeley
01 – 05/ 2015	University of California at Berkeley, Department of EECS
04 – 08/ 2005	Indian Institute of Science, Department of ECE
01 – 04/ 2005	Institut Eurecom, Sophia Antipolis, France
08 – 12/ 2004	University of Illinois at Urbana Champaign, Department of ECE

Education

1994 – 1998	Ph.D., Georgia Institute of Technology in Electrical Engineering.
1992 – 1994	M.S., Iowa State University in Electrical Engineering.
1988 – 1992	B.E., Coimbatore Institute of Technology , Electronics & Communications Engg.

Research Areas

- Artificial intelligence, Wireless Networks, and AI-Native Wireless Networks
- Physical layer/Medium access control layer technologies for 6G cellular
- Data Science (Algorithms for large-dimensional inference, sparse recovery)
- Coding theory, information theory, signal processing
- Data storage including emerging paradigms

Teaching Related Activities

- Service
 - Creator of Sprint Courses Initiative at the College of Engineering, 2023
 - PI for Generative AI Literacy grant from OpenAI which led to the establishment of the Generative AI Literacy Initiative within TAMIDS, 2025
 - Served on the Curriculum Committee for B.S. in Data Engineering Program
- Courses developed
 - Data Analytics II (Deep Learning) - for the Data Engineering Undergraduate Program
 - Probabilistic Graphical Models and Inference
 - Advanced topics in channel coding - a graduate course on the fundamentals of codes on graphs and iterative decoding, redesigned in 2022 with emphasis on modern applications
 - Wireless Communications - upgraded the graduate level wireless communications course in 2021 with emphasis on recent developments
- Courses taught - Data Science, Mathematics for Signal Processing, Channel coding for communications, Advanced Channel Coding, Information theory, Wireless communications, Digital Communications, Signals and Systems

Student Supervision

- Graduated Students - 20 Ph.D., 4 Ph.D. students were co-advised with other professors, 11 M.S. students
- Current Students - 4 Ph.D

Editorial Positions

- Guest editor, Journal of Communications Networks, Special issue on Communications and Networking Approaches to Combating COVID-19
- Associate Editor, Coding Techniques, IEEE Transactions on Information Theory, 2015-2018
- Area Editor, Coding Theory and Applications, IEEE Transactions on communications, 2007 - 2011
- Editor, IEEE Transactions on communications, 2005 - 2011
- Editor, IEEE Transactions on Wireless Communications, 2001 - 2005
- Associate editor for IEEE communications letters, 2000 - 2003

Honors

- Presidential Professor for Teaching Excellence, Texas A&M University
- Fellow of IEEE
- Inaugural member of the Academy of Eminent Educators (AE2) in the College of Engineering, Texas A&M University
- Recipient of the Best Paper Award from the 2025 Signal Processing, Artificial Intelligence, and Wireless Communications Conference
- Recipient of the 2022 IEEE Communication Society and Information Theory Society joint paper award

- Recipient of the 2020 IEEE Data Storage Technical Committee Best Paper Award
- Genesis award from the College of Engineering, Texas A&M University, 2023
- Inaugural Fellow of Institute of Engineering Education, Texas A&M University, 2018
- Best paper award at the International Conference on Vehicular Electronics and Safety, 2018
- Association of Former Students Distinguished Award for Teaching at the University level, 2018
- Outstanding Professor award, Department of Electrical & Computer Engg, Texas A&M University, Dec 2015
- Dean's excellence award, College of Engineering at Texas A&M University, May 2015
- Recipient of the Professional Progress in Engineering (Alumni award given to outstanding alumni under the age of 46) from Iowa State University, Ames, IA, April 2014
- Recipient of the Association of Former Students teaching award, Texas A&M University, 2012
- Recipient of the 2011-2012 William Keeler fellow award from the College of Engineering, Texas A&M University.
- Recipient of 2009 Halliburton fellowship from the College of Engineering, Texas A&M University.
- Recipient of the 2007 IEEE Data Storage Technical Committee Best Paper Award
- Outstanding Professor Award, Department of Electrical Engineering, Texas A&M University, 2002
- Faculty early CAREER development award, National Science Foundation, 2001
- Select Young Faculty Award within College of Engineering at Texas A&M University, 2001
- Nominated for Montague Teaching Excellence Award at Texas A&M University from Dept. of Electrical Engineering in Year 2001 (did not receive the award)
- Recipient of Harpole, Palmer and Litton Graduate Fellowship from the Dept. of Electrical Engineering, Iowa State University, Ames, IA
- Recipient of the Premium for Academic Excellence award from Graduate College, Iowa State University, Ames, IA
- First ranked student in the ECE department in B.E.

Keynotes, Plenary Lectures, and Major Tutorials

- Upcoming, Keynote talk at International Symposium on Information Theory and its Applications, November 2026, Kyoto, Japan
- Revisiting Compression and Communication through the Lens of Generative AI, Plenary talk at IEEE Information Theory and its Applications (ITA) workshop, 2026
- Department level Distinguished Lecture at the ECE Department at The University of Arizona, November 2025
- Revisiting Compression and Communication through the Lens of Generative AI, Keynote talk at IEEE Workshop on Signal Processing and AI in Wireless Communications, SPAWC, June 7-10, 2025
- Transformers for Compression and Estimation, Keynote talk at The 2024 Wireless and Optical Communications (WOCC) Conference, Taiwan, 2024
- Transformers for Compression and Estimation, Plenary talk at The 2024 International Conference on Signal Processing and Communications (SPCOM'24), Indian Institute of Science

- Reengineering the Uplink for Next-Generation Massive Multiple Access, Keynote talk at The Workshop on Random Access at IEEE WCNC'2024
- "Unsourced Multiple Access (UMAC): Information Theory and Coding", International Symposium on Information Theory, 2021
- "Algorithms for Coded Random Access and Inference in Large Dimensional Spaces – Bridging Fundamental Limits and Practical Schemes", SPCOM, 2020
- "The Peeling Decoder : Theory and Applications", Tutorial presented at the 2018 East Asian School on Information Theory, National Taiwan University, August 2018
- "The Peeling Decoder : Theory and Applications", Tutorial presented at the 2017 Australian School on Information Theory, Australian National University, Jan 18, 2017
- "The Peeling Decoder : Theory and Applications", Tutorial presented at the North American School of Information Theory, Duke University, June 21, 2016
- Key note talk at the 2016 ICC workshop on Massive Uncoordinated Multiple Access
- EXIT Charts: Properties, Extensions and Applications, Indian Institute of Science, 2005
- Low Density Parity Check Codes, 2005, IEEE Web-based tutorial

Externally Sponsored Research

- 1 PI-K.R. Narayanan, Co-PI - C. Tian, "NSF ECCS: Neural Joint Source-Channel Coding: The Interplay Between Theory and Practice", National Science Foundation \$450,000.
- 2 PI - I-Hong Hou, Co-PIs - D. Kalathil, P.R. Kumar, and K.R. Narayanan, "RAINS: Resilient AI with in-Network and Edge Synergism", DEVCOM, \$1,212,739
- 3 PI-K.R. Narayanan, Co-PIs J.-F. Chamberland, S. Hoyos, S. Palasole, "RINGS: Resilient Wireless Systems for Future Uplink Traffic through Cell-Free, Loosely Coordinated Access", National Science Foundation \$1,00,000, 05/01/2022-04/30/2025
- 4 PI-J.-F. Chamberland, Co-PIs K.R. Narayanan, Y. Polyanskiy, "Collaborative Research: CIF: Small: Low-Complexity Algorithms for Unsourced Multiple Access and Compressed Sensing in Large Dimensions", National Science Foundation \$495,000, 10/01/2021-09/30/2024
- 5 PI-K.R. Narayanan, Co-PI A. Heidarzadeh, "Numerically-Stable Large-Scale Coded Distributed Computing", National Science Foundation, \$499,015, 07/01/2020-06/30/2023
- 6 PI-P.R. Kumar, Co-PI - J.-F. Chamberland, K.R. Narayanan, S. Shakkottai, "SII Planning: Theory, Practice and Reality of Spectrum", National Science Foundation, \$300,000, 08/15/2020-07/31/2021
- 7 PI - K. R. Narayanan, Co-PI - A. Heidarzadeh, "RAPID: Accelerated Testing for COVID-19 using Group Testing", National Science Foundation, \$114,162, 05/01/2020-04/31/2021
- 8 PI - N. Duffied, Co-PIs - K.R. Narayanan, S. Shakkottai, A. Talebpour, "EAGER: Real-Time: Learning-Mediated Control for Traffic Shaping", National Science Foundation, \$300,000, 10/01/2018-09/30/2020
- 9 PI - K.R. Narayanan, Qualcomm faculty award, \$75,000, 2018
- 10 PI - K.R. Narayanan, Qualcomm faculty award, \$75,000, 2017
- 11 PI - A. Jiang, Co-PIs - K.R. Narayanan and J. Bruck, "CIF: Small: Collaborative Research: Error Correction with Natural Redundancy", National Science Foundation, \$299,999 (only the TEES portion), Oct 1 2017 - Sept 31, 2020.

- 12 PI - K.R. Narayanan, Co-PIs - H.D. Pfister and G. Choi, "Collaborative Research: Advanced Coding Techniques for Next-Generation Optical Communications, National Science Foundation, \$366,765, Sept. 15, 2016 - Sept 14, 2019
- 13 PI - K.R. Narayanan, Co-PIs - J.-F. Chamberland and P.R. Kumar, "CIF Small: Massive Uncoordinated and Sporadic Multiple Access - Strengthening Connections between Coding and Random Access", National Science Foundation, \$495,010, June 15, 2016 - June 14, 2019
- 14 PI - K.R. Narayanan, Co-PIs - A. Karsilayan, E. Serpedin and J. Silva-Martinez, "EARS: Enhancing Radio-Frequency Spectrum Through Interference Resilient Cognitive Radio Systems: Design, Performance Analysis and Optimization", National Science Foundation, \$407,334, October 1, 2015 - September 31, 2018
- 15 PI - A. L. N. Reddy, Co-PI K.R. Narayanan and P. Gratz, "I/URC Phase II: Center on Intelligent Storage", National Science Foundation, \$300,000, September 15, 2014 - August 31, 2019
- 16 PI - M. Smotherman, Senior Personnel - K.R. Narayanan, "Networking Strategies used by Bats to Improve Social Sonar", National Science Foundation, \$660,000 (pro-rated amount \$30,433), August 15, 2014 to August 14, 2017
- 17 PI - B. Nazer, Co-PI - K. R. Narayanan (TEES PI) and B. Aazhang, "CIF: Medium: Collaborative Research: Interference-Aware Cooperation via Structured Codes", National Science Foundation, \$1,086,558, May 1, 2013 to April 1, 2017
- 18 PI - H. Pfister, Co-PI - K.R. Narayanan, "CIF:Small: Design and Analysis of Spatially-Coupled Coding Systems", National Science Foundation, \$494,231, August 1, 2013 to July 31, 2016
- 19 PI - J. Boutros, Co-PI - K.R. Narayanan and E. Viterbo, "Information Processing via Lattices in Digital Data Networks", Qatar National Research Foundation, \$1,045,374, October 15, 2012 - October 14, 2015
- 20 PI - A. Ruimi, Co-PI - K. R. Narayanan, M. Fratarcangeli, B. Nour, J. N. Reddy and A. R. Srinivasa, "Surgical Thread Simulation Using a Novel Information-Theory Approach", Qatar National Research Foundation, \$1,040,321, October 15, 2012 - October 14, 2015
- 21 PI - K.R. Narayanan, co-PI - H.D. Pfister, "Coding for Fiber-Optic Communication", Xilinx Inc, \$100,000, Sept 1, 2011 - May 31, 2013
- 22 PI - K.R. Rajagopal, Co-PI's - A.R. Srinivasa, J. Criscione and K.R. Narayanan, "CDI Type II/Collaborative Research: A New Approach to the Modeling of Clot Formation and Lysis in Arteries", National Science Foundation, \$875,165, (Pro-rated amount - \$214,114), Sept 1, 2010 - Aug 31, 2013
- 23 PI - J.F. Chamberland, Co-PI's - H.D. Pfister and K.R. Narayanan, "Fundamental Limits on Delay Constrained Communications", National Science Foundation, \$ 300,000, Sept 1, 2008 - Aug 31, 2011
- 24 PI - X. Zhang, Co-PIs - K.R. Narayanan (TEES PI), H. D. Pfister, "GOALI : Advanced Coding and Signal Processing for Magnetic Recording: From Theory to Implementation", National Science Foundation, \$175,000 (dollar amount awarded to Texas A&M University), May 1 2008 - April 31 2011
- 25 PI - M.S. Alouini, Co-PIs - K.R. Narayanan, H. D. Pfister, and K. Qaraqe, "Universal Signaling Schemes for Multimedia Transmission over Wireless Networks", Qatar National Research Foundation, \$575,100, January 2008-December 2010
- 26 PI - K.R. Narayanan, "Joint Source-Channel Coding for Wireless Networks", National Science Foundation, \$300,000, September 2007 - August 2010
- 27 PI - A. R. Srinivasa, Co-PIs - J. Froyd, A. Jayaraman, D. Maxwell, K.R. Narayanan, "CCLI Phase I: An Engineering Emphasis for Preparing Students for First-year Engineering Curricula", National Science Foundation, \$200,000, June 2009 - May 2011

- 28 PI - K.R. Narayanan, Co-PI H. D. Pfister, "Soft Decision Decoding of Reed Solomon Codes for High Density Recording Channels", International Storage Industry Consortium, \$32,000, January 2007 - December 2007
- 29 PI - K.R. Narayanan, "Understanding and Mitigating Error Floors in Iterative Decoding with Applications to Magnetic Recording", International Storage Industry Consortium, \$30,000, Jan 2006 - December 2006
- 30 PI - K.R. Narayanan, "Universal Source and Channel Codes for Non-Ergodic Channels", National Science Foundation, \$189,586, Sept. 2005 - Aug 2008
- 31 PI - K.R. Narayanan, "Low Complexity Iterative Decoding for Digital Magnetic Recording", Seagate Research Inc, Pittsburgh, \$75,106, 01/04-12/05.
- 32 PI - K.R. Narayanan, "Coding for Long Haul Optical Networks", Azea Networks, United Kingdom, \$45,000, Jan 01, 2004 to Dec 31, 2004
- 33 PI - K.R. Narayanan, CAREER award "Design and Analysis of Iteratively Decodable Codes for Wireless Communications and Digital Magnetic Recording", National Science Foundation, \$300,000, Feb 01, 2001 - Jan 31, 2007
- 34 PI - K. R. Narayanan, "Low Density Parity Check Codes and Turbo-Like Codes for Wireless Communications", National Science Foundation, \$179,925, 09/01/2000 - 08/31/2003
- 35 PI - K.R. Narayanan, "Advanced Coding Techniques and Low Complexity Decoding for Wireless Data Networks", Advanced Technology Program, Texas Higher Education coordinating Board, \$72,600, 01/01/00-12/31/01
- 36 PI - K. R. Narayanan, " Low Density Parity Check Codes for High Order Signal Constellations", Tadiran Microwave Networks, \$38,000, 06/01/00-12/01/01
- 37 PI - K.R. Narayanan, "Low Complexity Iterative Decoding for Digital Magnetic Recording", Seagate Research Inc, Pittsburgh, \$80,106, 01/02-12/03
- 38 PI - K. R. Narayanan, "Concatenated Coding and Iterative Decoding", Rockwell Collins University Grants Program, \$41,500, 09/01/00-05/01/02
- 39 PI - K.R. Narayanan, "Research in Turbo Coding", GS Labs, \$16,000
- 40 PI - K. R. Narayanan, "Low Density Parity Check Codes for Wireless Communications", Texas Instruments, \$20,000, 09/01/00-08/31/01
- 41 PIs - C.N.Georgiades, S.Miller, K.Narayanan, & X. Wang, "Space-Time Processing", Motorola Inc, Austin, \$60,000, 01/01/2000-01/01/2001
- 42 PI - G. Choi, Co-PI - K.R. Narayanan, "ASIC Design and Implementation of Advanced Coding Techniques for Wireless Data Networks", Advanced Technology Program, Texas Higher Education coordinating Board, \$142,600, 01/01/02-12/31/03

Book Chapters

- 1 Y. Wang, Y.-C. Huang, A. G. Burr, and K.R. Narayanan, Multilevel Lattices for Compute-and-Forward and Lattice Network Coding, pp. 201-240, in Number Theory Meets Wireless Communication, Springer 2020
- 2 K.R. Narayanan, Turbo Equalization, in Wiley Encyclopedia on Telecommunications, Editor - J.G. Proakis, Wiley, 2003

- 3 J. Li, E. Kurtas and K.R. Narayanan, Turbo Product Codes for Partial Response Channels, in CRC Handbook, CRC Press

Selected Research Publications

- 1 Ankireddy, S. K., Narayanan, K., & Kim, H. (2025). Lightcode: Light analytical and neural codes for channels with feedback. *IEEE Journal of Selected Areas in Communication*, 43(1), 1230–1244.
- 2 Ebert, J. R., Chamberland, J.-F., & Narayanan, K. R. (2025). Sparse regression LDPC codes. *IEEE Transactions on Information Theory*, 73(1), 167–191.
- 3 Rajagopalan, V., Kunde, V. T., Valmeekam, C. S. K., Narayanan, K., Shakkottai, S., Kalathil, D., & Chamberland, J.-F. (2025). Transformers are provably optimal in-context estimators for wireless communications. *AISTATS 2025*. Also available at <https://arxiv.org/abs/2311.00226>.
- 4 Mattu, S. R., Khan, I. A., Khammammetti, V., Dabak, B., Mohammed, S. K., Narayanan, K., & Calderbank, R. (2024). Delay-doppler signal processing with zadoff-chu sequences. *arXiv preprint arXiv:2412.04295*.
- 5 Valmeekam, C. S. K., Narayanan, K., Kalathil, D., Chamberland, J.-F., & Shakkottai, S. (2023). LLMZip: Lossless text compression using large language models. *arXiv preprint arXiv:2306.04050*.
- 6 Amalladinne, V. K., Ebert, J. R., Chamberland, J.-F., & Narayanan, K. R. (2022). An enhanced decoding algorithm for coded compressed sensing with applications to unsourced random access. *Sensors*, 22(2), 676.
- 7 Ebert, J. R., Amalladinne, V. K., Rini, S., Chamberland, J.-F., & Narayanan, K. R. (2022). Coded demixing for unsourced random access. *IEEE Transactions on Signal Processing*, 70, 2972–2984.
- 8 Gkagkos, M., Narayanan, K. R., Chamberland, J.-F., & Georghiades, C. N. (2022). FASURA: A scheme for quasi-static fading unsourced random access channels. *to appear in IEEE Transactions on Communications*, *arXiv preprint arXiv:2202.11042*.
- 9 Pradhan, A. K., Amalladinne, V. K., Vem, A., Narayanan, K. R., & Chamberland, J.-F. (2022). Sparse IDMA: A joint graph-based coding scheme for unsourced random access. *IEEE Transactions on Communications*, 70(11), 7124–7133.
- 10 Xuan, Z., & Narayanan, K. (2022). Low-delay analog joint source-channel coding with deep learning. *IEEE Transactions on Communications*, 71(1), 40–51.
- 11 Amalladinne, V. K., Pradhan, A. K., Rush, C., Chamberland, J.-F., & Narayanan, K. R. (2021). Unsourced random access with coded compressed sensing: Integrating AMP and belief propagation. *IEEE Transactions on Information Theory*, 68(4), 2384–2409.
- 12 Pradhan, A. K., Heidarzadeh, A., & Narayanan, K. R. (2021). Factored LT and factored Raptor codes for large-scale distributed matrix multiplication. *IEEE Journal on Selected Areas in Information Theory*, 2(3), 893–906.
- 13 Soma, D. K., Pradhan, A. K., & Narayanan, K. (2021). Errors and erasures decoding of product codes for optical transport networks. *IEEE Communications Letters*, 25(8), 2482–2486.
- 14 Amalladinne, V. K., Chamberland, J.-F., & Narayanan, K. R. (2020). A coded compressed sensing scheme for unsourced multiple access. *IEEE Transactions on Information Theory*, 66(10), 6509–6533.
- 15 Hasanzadeh, A., Hajiramezanali, E., Boluki, S., Zhou, M., Duffield, N., Narayanan, K., & Qian, X. (2020). Bayesian graph neural networks with adaptive connection sampling. In *International conference on machine learning (ICML)* (pp. 4094–4104). PMLR.
- 16 Pradhan, A. K., Amalladinne, V. K., Narayanan, K. R., & Chamberland, J.-F. (2020). Polar coding and random spreading for unsourced multiple access. In *Icc 2020-2020 ieee international conference on communications (ICC)* (pp. 1–6). IEEE.

- 17 Hajiramezanali, E., Hasanzadeh, A., Narayanan, K., Duffield, N., Zhou, M., & Qian, X. (2019). Variational graph recurrent neural networks. *Advances in neural information processing systems (Neurips)*, 32.
- 18 Hasanzadeh, A., Hajiramezanali, E., Narayanan, K., Duffield, N., Zhou, M., & Qian, X. (2019). Semi-implicit graph variational auto-encoders. *Advances in neural information processing systems (Neurips)*, 32.
- 19 Subramaniam, A. M., Heidarzadeh, A., & Narayanan, K. R. (2019a). Collaborative decoding of polynomial codes for distributed computation. In *2019 IEEE information theory workshop (ITW)* (pp. 1–5). IEEE.
- 20 Subramaniam, A. M., Heidarzadeh, A., & Narayanan, K. R. (2019b). Random Khatri-Rao-product codes for numerically-stable distributed matrix multiplication. In *2019 57th Annual Allerton Conference on Communication, Control, and Computing (Allerton)* (pp. 253–259). IEEE.
- 21 Vem, A., Narayanan, K. R., Chamberland, J.-F., & Cheng, J. (2019). A user-independent successive interference cancellation based coding scheme for the unsourced random access Gaussian channel. *IEEE Transactions on Communications*, 67(12), 8258–8272.
- 22 Huang, Y.-C., Narayanan, K. R., & Wang, P.-C. (2018). Lattices over algebraic integers with an application to compute-and-forward. *IEEE Transactions on Information Theory*, 64(10), 6863–6877.
- 23 Kim, D., Narayanan, K. R., & Ha, J. (2018). Symmetric block-wise concatenated BCH codes for NAND flash memories. *IEEE Transactions on Communications*, 66(10), 4365–4380.
- 24 Kong, H., Cheng, J., Narayanan, K., & Hu, J. (2018). DUCER: A fast and lightweight error correction scheme for in-vehicle network communication. In *2018 IEEE International Conference on Vehicular Electronics and Safety (ICVES)* (pp. 1–6). IEEE.
- 25 Huang, Y.-C., & Narayanan, K. R. (2017). Construction π_A and π_D lattices: Construction, goodness, and decoding algorithms. *IEEE Transactions on Information Theory*, 63(9), 5718–5733.
- 26 Arumugam, J., Bukkapatnam, S. T., Narayanan, K. R., & Srinivasa, A. R. (2016). Random forests are able to identify differences in clotting dynamics from kinetic models of thrombin generation. *PloS one*, 11(5), e0153776.
- 27 Tunali, N. E., Huang, Y.-C., Boutros, J. J., & Narayanan, K. R. (2015). Lattices over Eisenstein integers for compute-and-forward. *IEEE Transactions on Information Theory*, 61(10), 5306–5321.
- 28 Wang, Y., Narayanan, K. R., & Huang, Y.-C. (2015). Interleaved concatenations of polar codes with BCH and convolutional codes. *IEEE Journal on Selected Areas in Communications*, 34(2), 267–277.
- 29 Wang, P.-C., Huang, Y.-C., & Narayanan, K. R. (2014). Asynchronous physical-layer network coding with quasi-cyclic codes. *IEEE Journal on Selected Areas in Communications*, 33(2), 309–322.
- 30 Hern, B., & Narayanan, K. R. (2013). Multilevel coding schemes for compute-and-forward with flexible decoding. *IEEE transactions on information theory*, 59(11), 7613–7631.
- 31 Huang, Y.-C., Tunali, N. E., & Narayanan, K. R. (2013). A compute-and-forward scheme for Gaussian bi-directional relaying with inter-symbol interference. *IEEE transactions on communications*, 61(3), 1011–1019.
- 32 Doraiswamy, S., Narayanan, K. R., & Srinivasa, A. R. (2012). Finding minimum energy configurations for constrained beam buckling problems using the Viterbi algorithm. *International Journal of Solids and Structures*, 49(2), 289–297.
- 33 Huang, Y.-C., & Narayanan, K. R. (2012). Joint source-channel coding with correlated interference. *IEEE transactions on communications*, 60(5), 1315–1327.
- 34 Narayanan, K., & Srinivasa, A. (2012). Shannon-entropy-based nonequilibrium “entropic” temperature of a general distribution. *Physical Review E*, 85(3), 031151.

- 35 Narayanan, K. R., & Pfister, H. D. (2012). Iterative collision resolution for slotted aloha: An optimal uncoordinated transmission policy. In *2012 7th international symposium on turbo codes and iterative information processing (istc)* (pp. 136–139). IEEE.
- 36 Parag, P., Chamberland, J.-F., Pfister, H. D., & Narayanan, K. (2012). Code-rate selection, queueing behavior, and the correlated erasure channel. *IEEE transactions on information theory*, 59(1), 397–407.
- 37 Nguyen, P. S., Pfister, H. D., & Narayanan, K. R. (2011). On multiple decoding attempts for reed–solomon codes: A rate-distortion approach. *IEEE transactions on information theory*, 57(2), 668–691.
- 38 Wilson, M. P., Narayanan, K., Pfister, H. D., & Sprintson, A. (2010). Joint physical layer coding and network coding for bidirectional relaying. *IEEE Transactions on Information Theory*, 56(11), 5641–5654.
- 39 Narayanan, K. R., Jiang, J., & Nangare, N. A. (2009). Iterative decoding of linear block codes by adapting the parity check matrix. US Patent 7,519,898.
- 40 Bhattad, K., & Narayanan, K. R. (2009). A note on the rate of decay of mean-squared error with snr for the awgn channel. *IEEE transactions on information theory*, 56(1), 332–335.
- 41 Narayanan, K. R., & Srinivasa, A. R. (2009). Using discrete optimization algorithms to find minimum energy configurations of slender cantilever beams with non-convex energy functions. *Mechanics Research Communications*, 36(7), 811–817.
- 42 Bhattad, K., Narayanan, K. R., & Caire, G. (2008). On the distortion snr exponent of some layered transmission schemes. *IEEE transactions on information theory*, 54(7), 2943–2958.
- 43 Jiang, J., & Narayanan, K. R. (2008). Algebraic soft-decision decoding of reed–solomon codes using bit-level soft information. *IEEE transactions on information theory*, 54(9), 3907–3928.
- 44 Caire, G., & Narayanan, K. (2007). On the distortion snr exponent of hybrid digital–analog space–time coding. *IEEE Transactions on Information Theory*, 53(8), 2867–2878.
- 45 Bhattad, K., & Narayanan, K. R. (2006a). A decision feedback based scheme for slepian-wolf coding of sources with hidden markov correlation. *IEEE communications letters*, 10(5), 378–380.
- 46 Bhattad, K., & Narayanan, K. R. (2006b). An mse-based transfer chart for analyzing iterative decoding schemes using a gaussian approximation. *IEEE transactions on information theory*, 53(1), 22–38.
- 47 Jiang, J., & Narayanan, K. R. (2006). Iterative soft-input soft-output decoding of reed–solomon codes by adapting the parity-check matrix. *IEEE Transactions on Information Theory*, 52(8), 3746–3756.
- 48 Lan, C.-F., Xiong, Z., & Narayanan, K. R. (2006). Source-optimized irregular repeat accumulate codes with inherent unequal error protection capabilities and their application to scalable image transmission. *IEEE transactions on image processing*, 15(7), 1740–1750.
- 49 Peiris, B., Narayanan, K., & Miller, S. L. (2006). A spectral domain approach to design spreading sequences for ds-cdma systems in frequency selective fading channels. *IEEE transactions on wireless communications*, 5(9), 2386–2395.
- 50 Bhattad, K., Narayanan, K. R. et al. (2005). Weakly secure network coding. *NetCod, Apr*, 104.
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- 71 Narayanan, K. R. (2001). Effect of precoding on the convergence of turbo equalization for partial response channels. *IEEE Journal on Selected Areas in Communications*, 19(4), 686–698.

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Patents

- 1 K.R. Narayanan, J. Jiang and N. Nangare, “Iterative decoding of linear block codes by adapting the parity check matrix”, U.S. Patent 7519898, April 2009
- 2 H.-J. Tarn, K.R. Narayanan, R.M. Rao, R. N. Mazareh, “Circuitry and method for forward error correction”, U.S. Patent 9009577B1, April 2015
- 3 H.D. Pfister, K.R. Narayanan, R. Mazahreh, R. M. Rao, “Forward Error Correction”, U.S. Patent 8959418B1
- 4 N. E. Tunali, R. M. Rao, R. N. Mazahreh, K. R. Narayanan, “Matrix Expansion”, U. S. Patent US9203440B1, 2015

Invited Talks

- 1 Transformers for Compression and Estimation, IEEE Communication Theory Workshop, May 2024
- 2 Transformers for Compression and Estimation, IEEE Communication Theory Workshop, Aira Technologies, April 2024
- 3 Transformers for Compression and Estimation, IEEE Communication Theory Workshop, Intel Labs seminar, January 2024
- 4 “On the Advantages of Asynchrony in Unsourced Random Access”, International Symposium on Topics in Coding, Brest, France, 2023
- 5 “Coding, Signal Processing and Multiple Access for massive Machine Type Communication”, 2+1 event, IEEE Communication Society
- 6 “Coding, Signal Processing and Multiple Access for massive Machine Type Communication”, Next Generation Alliance, Nov. 2022
- 7 “Connections between Compressed Sensing and Multiple Access Communication”, UT Austin Wireless Seminar, Nov. 2022
- 8 “Connections between Compressed Sensing and Multiple Access Communication”, Center for Networked Intelligence Seminar, Jan 2022

- 9 "Coding for Distributed Computing", Department Colloquium, North Carolina State University, 2021
- 10 "Coded Compressed Sensing: A Sub-linear Time Algorithm for Compressed Sensing", SIAM conference on Data Science, July 2020
- 11 "Erasure-based Soft-Decision Decoding of Product Codes", Oberpfaffenhofen workshop on coding, Feb 2019
- 12 "A Sub-linear Time Algorithm for Sub-string Matching", Conference on Science of Electrical Engineering, Eilat, Israel, December 2018
- 13 "A Couple Compressive Sensing Scheme for Unsourced Multiple Access", Coding, Cooperation and Security workshop, Technion, Haifa, December, 2018
- 14 "A Sub-linear Time Algorithm for Sub-string Matching", NSF Serbia joint workshop on Data Science, Belgrade, August 2018
- 15 "The Peeling Decoder : Theory and Applications", Tutorial presented at the 2018 East Asian School on Information Theory, August 2018
- 16 "An Iterative Decoding Scheme for Unsourced Multiple Access", NSF work on multiple access, MIT, Nov. 2017
- 17 "The Peeling Decoder : Theory and Applications", Tutorial presented at the 2017 Australian School on Information Theory, Australian National University, Jan 18, 2017
- 18 "The Peeling Decoder : Theory and Applications", Tutorial presented at the North American School of Information Theory, Duke University, June 21, 2016
- 19 "Exploring Connections between Sparse Fourier Transform Computation and Decoding of Product Codes", Simons Institute, Information Theory Program Reunion, June 2016
- 20 "Coding-Inspired Massive Multiple Access for Next-Generation Wireless Networks", Norwegian University of Science and Technology
- 21 "Coding-Inspired Massive Multiple Access for Next-Generation Wireless Networks", Samsung Research, Dallas, December 2015
- 22 "Uncoordinated Massive Multiple Access", Chinacom, August 2015
- 23 "Half Product Codes for Flash Memory", Intel, April 2015
- 24 "Half Product Codes for Flash Memory", Hitachi, San Jose, March 2015
- 25 "Uncoordinated Massive Multiple Access", Information Theory at Its Applications Workshop, San Diego, Feb 2015
- 26 "Lattices from Codes for Harnessing Interference : An Overview and Some Generalizations", IEEE Information Theory Workshop, Tasmania, Australia, Nov. 2014
- 27 "Coding and Modulation for Physical Layer Network Coding", Signal Processing and Communications Conference, Indian Institute of Science, July 2014
- 28 "Multilevel Lattice Codes for the Interference Channel", IEEE Communication Theory Workshop, May 2014
- 29 "Physical Layer Network Coding", ECE Seminar, Iowa State University, April 2014
- 30 "Lattice Codes based on Product Constructions", Information Theory and its Applications Workshop, Feb 2014

- 31 "Lattices based on Eisenstein Integers for Compute and Forward", *50th Allerton Conference on Communications, Control and Computing*, 2012
- 32 "Codes based on Eisenstein Integers for Compute and Forward", Information Theory and Applications Workshop, San Diego, 2012
- 33 "Coding for Compute and Forward", Banff International Research Station (BIRS) workshop on Algebraic Structure in Network Information Theory, August 2011
- 34 "Coding for Compute and Forward", Signal Processing and Communications Conference SPCOM 2010, University of Hawaii, November 2010
- 35 "Coding for Compute and Forward", Signal Processing and Communications Conference SPCOM 2010, Indian Institute of Science, Bangalore, July 2010
- 36 "Coding for Compute and Forward", Indian Institute of Technology, Madras, July 2010
- 37 "Joint Physical Layer and Network Layer Coding for Bi-Directional Relaying", Department of Electrical and Computer Engineering, Texas A&M University, Doha, Qatar, May 2010
- 38 "Joint Physical Layer and Network Layer Coding for Bi-Directional Relaying with Asymmetric Links", Information Theory and its Applications workshop, University of California at San Diego, La Jolla, CA, January 2009
- 39 "Joint Physical Layer and Network Layer Coding for Bi-Directional Relaying", Department of Electrical and Computer Engineering, University of Massachusetts, Amherst, November 2008
- 40 "Joint Physical Layer and Network Layer Coding for Bi-Directional Relaying", Department of Electrical and Computer Engineering, University of Arizona, Tucson AZ, February 2008
- 41 "On the Thermodynamic Temperature of a General Distribution", Fisher Information Workshop, Department of Optics, University of Arizona, Tucson, AZ, February 2008
- 42 "Joint Source Channel Coding for Non-Ergodic MIMO Channels", WINLAB Seminar, Rutgers University, April 24, 2007
- 43 "Joint Source Channel Coding for Non-Ergodic MIMO Channels", Columbia University, April 23, 2007
- 44 "Joint Source Channel Coding for Non-Ergodic MIMO Channels", MIT LIDS Colloquium, April 3, 2007
- 45 "Hybrid Digital Analog Coding with Side Information", UCSD Workshop on Information Theory and its Applications, February 2007
- 46 "Bit Level Soft Decision Decoding of Reed Solomon Codes", American Mathematical Society meeting, New Orleans, LA, January 2007
- 47 "Hybrid Digital Analog Strategies for Broadcasting", Rice University, TX, October 2006
- 48 "Iterative Signal Processing versus Decision Feedback Signal Processing", ICCSS, March 2006
- 49 "Hybrid Digital and Analog Space-Time Coding", The University of Texas at Austin, Feb 2006
- 50 "Further Results on the SNR Exponent of Hybrid Digital Analog Space-Time Codes", UCSD Workshop on Information theory, Feb 2006
- 51 "On the SNR Exponent of Hybrid Digital Analog Space-Time Codes", Indian Institute of Science, Bangalore, India, December 2005
- 52 "Soft Decision Decoding of RS Codes", Indian Institute of Science, Bangalore, India, June 2005 and again in December 2005 with updated results.

- 53 "To Iterate or Feedback? How to Achieve Universally Good Performance on Channels with Memory", Indian Institute of Science, Bangalore, India, June 2005.
- 54 "To Iterate or Feedback? How to Achieve Universally Good Performance on Channels with Memory", Indian Institute of Technology, Madras, India, June 2005.
- 55 "To Iterate or Feedback? How to Achieve Universally Good Performance on Channels with Memory", Georgia Institute of Technology, Lorraine, Metz, France, April 2005
- 56 "Soft Decision Decoding of RS Codes", Ecole Nationale Supérieure des Telecommunications (ENST) - Paris, France, April 2005
- 57 "To Iterate or Feedback? How to Achieve Universally Good Performance on Channels with Memory", Ecole Nationale Supérieure des Telecommunications (ENST - Paris), Paris, France, April 2005.
- 58 "Soft Decision Decoding of RS Codes", Ecole Nationale Supérieure des Telecommunications (ENST) - Bretagne, France, March 2005
- 59 "To Iterate or Feedback? How to Achieve Universally Good Performance on Channels with Memory", Institut Eurecom, Sophia Antipolis, France, Jan 2005.
- 60 "Soft Decision Decoding of RS Codes", University of Notre Dame, IN, November 2005
- 61 "To Iterate or Feedback? How to Achieve Universally Good Performance on Channels with Memory", The University of Texas at Austin, October 2004
- 62 "Decision Feedback Signal Processing for Achieving Near Capacity Performance on Channels with Memory", The University of Illinois at Urbana-Champaign, October 2004
- 63 "Iterative Decoding of RS Codes Based on Adaptive Parity Check Matrices", March 2004, Workshop on Theoretical Advances in Information Recording, DIMACS, Rutgers University
- 64 "Simplified Design and Analysis of LDPC Codes for ISI Channels with Turbo Equalization", April 2003, Harvard University.
- 65 "Design of LDPC codes with MMSE Turbo Equalization", Information Theory Workshop, October 2002
- 66 "Product Accumulate Codes on Fading Channels", Asilomar Conference on Signals and Systems, November 2002
- 67 "Design and Analysis of Codes for Turbo Equalization with Optimal and Suboptimal Soft Output Decoding", Allerton conference on communications and control, October 2002 (upcoming)
- 68 "Product Accumulate Codes: Construction and Properties", Information Theory Workshop, September 2001
- 69 "Effect of Precoding on the Convergence of Turbo Equalization for Partial Response Channels", The Magnetic Recording Conference, Santa Clara, August 2000
- 70 "Design considerations for iterative equalization and decoding", Conference on Information Science and Systems (CISS), Princeton, Mar 2000
- 71 "Iterative demodulation, equalization and decoding", IEEE signal processing society, Dallas TX, January 20, 2000
- 72 "Performance of Turbo Coded CPM with Iterative Demodulation", IEEE Military Commun. Conf, 1999
- 73 "Turbo coding and Low density parity check codes for microwave communications", Tadiran Microwave Networks, Houston, TX, December 2, 1999

Consulting Experience

- (09/21 - Present) Bartlit Beck LLP, Chicago, IL
- (10/21 - 01/23) AiRANACULUS, Chemsford, MA
- (02/17 - 07/19) Warren Lex LLP, San Francisco, CA
- GLS Labs, Atlanta, GA
- Microwave Networks, Stafford, TX, 2004
- Wilmer Hale Associates LLP, Boston, MA
- SiRF Technologies, San Jose, CA
- sw4a Inc
- Xilinx Inc

Significant Professional Activities

- Co-organizer, Application-Driven Coding Theory Workshop at the Simons Institute for Theoretical Computing, March 2024
- Technical Program Co-chair, 2021 IEEE Information Theory Workshop
- Technical Program Co-chair, 2018 IEEE Information Theory Workshop
- Technical Program Co-chair, 2010 IEEE International Symposium on Information Theory
- Member, Board of Governors, IEEE Information Theory Society, 2016-2017
- Lead guest editor, IEEE Journal of Selected Areas on Communications special issue on "Equalization with Applications to Wireless Communications", February 2008
- General Co-Chair, IEEE Information Theory Workshop 2007 in Lake Tahoe, CA, September 2007
- Ph.D. thesis reviewer for Aalto University, the Technical University of Berlin, Indian Institute of Science, Australia National University, Tel Aviv University, University of Toronto, Lund University, University of Oulu, Norwegian University of Science and Technology
- Proposal reviewer and/or panelist for the National Science Foundation, Israel Ministry of Science, MICRO proposals for state of California, State of S. Carolina, Hongkong ministry of science, SABIC proposals for King Fahd University of Petroleum and Mines, Saudi Arabia, NSERC in Canada
- Member of best student paper award committee, IEEE signal processing society, data storage sub committee, Member of best paper award committee IEEE Comsoc.